WEEK 6 · DAY 3 · RULE BOOK

Project Clinic and Demo Day

Follow it step by step. Every common error and its fix is here.

Your goal today: fix the last problems in your project at the clinic, deliver a **5-minute demo**, and hand in your report. In the notebook you rehearse the self-check on Ada's draft: **12 green ✅ ticks** to collect. Keep this page open next to Colab (or next to the training server).

Before you start (2 minutes)

- ☐ Open `W6D3-exercise` in Colab → **File** → **Save a copy in Drive**.
- ☐ Run **STEP 0**. Wait for ✅ `Ready!`
- ☐ Run the ▶️ **self-check toolkit** cell. Everything below needs it.
- ☐ Run every cell in order. Each ✍️ cell depends on the cells above it.

Real NIGCOMSAT data never goes into Colab, into an AI tool, or onto GitHub. Parts A–D use a synthetic course file. Part E (your own data) is for the **training server** only: **File** → **Download** → **Download .ipynb**, then open it there.

The exercise, step by step

Task	What you do	The tool you need
A1	Run the hygiene report on Ada's export	the frequency for "weekly, on Mondays": `"W-MON"`
A2	Count what it found	(duplicates, missing): read the two ⚠️ lines
A3	Fix the export	`GAP_FILL = "interpolate"` (and say so in your report)
A4	Fix the leaky 4-week mean	`.shift(1)` : week *t* only sees the 4 weeks before it
B1	Split by time	`CUT = "2025-11-03"` : the first test week
B2	The comparison table	`TERMINALS = train["active_vsat_terminals"]`
B3	Measure Ada's "90%" band	`coverage(test_y, fc - 3, fc + 3)`
B4	The honest band	the 90% quantile of past misses: `Q = 0.9`
C1	Name the report files by date	`"%Y-%m-%d"` : year first, so files sort by date
C2	Schedule it for 07:00 every day	`"0 7 * * *"`
D1	Write the 4-sentence recommendation	one number in the Observation

Task	What you do	The tool you need
D2	Write three honest limitations	misses · how late or how far · what you called normal

Your self-check toolkit (copy it into your project)

```
hygiene(df, "W-MON")                          # index, order, repeats, holes, frequency,
time zone
leak_test(lambda d: d["x"].shift(1), df)    # True = honest, False = it needed the
future
# fails loudly if the split is not by time, or the gap is shorter than the embargo:
check_temporal_split(train.index, test.index, embargo="24h")
compare(actual, {"base": f1, "mine": f2})   # same weeks, same scorer, best first
coverage(actual, lower, upper)              # about 0.9 for an honest 90% band
recommendation(situation, observation, implication, action)
```

Demo-day rules

#	Rule	Why
1	**5 minutes, then 1 question.** The timer is the MC.	Everyone presents inside the hour
2	**Six beats:** problem 0:45 · data 0:45 · method 0:45 · honest test 1:00 · the tool live 1:00 · recommendation 0:45	The honest test and the tool carry the most marks
3	**Show the tool live** from the training server, with a screenshot as backup	"Deployed" means it runs where your unit can use it
4	**One table** for the honest test: your method vs the baseline, same period, same scorer	Judges can't read three tables in a minute
5	**End on the recommendation**, not on "any questions?"	It's the sentence your unit head repeats
6	**Say one limitation out loud**	Unit heads trust the engineer who knows where the tool stops
7	**No real data on the projector** if outsiders are present	Show charts and summaries, not raw rows
8	**Two stars and a wish** for every other presenter	Kind, specific, about the work

Final pre-submission checklist

☐ **Agreement:** my unit head's email names the process, the data and who the tool is for. It's in my submission.

☐ **Hygiene:** DatetimeIndex, declared frequency, one time zone (written down), gaps and duplicates handled and **explained**.

☐ **Honest chart:** sensible axis, the threshold drawn, a title that states the finding.

☐ **Season:** the period comes from a correlogram, and I decomposed with it.

☐ **Leak test:** every feature passes. Trailing windows and positive shifts only.

☐ **Split by time** with an `assert` and an embargo ≥ my longest feature window.

☐ **Baseline** on the same period, with the same scorer.

☐ **Spread / repeated calls / coverage** reported (folds for forecasts, several origins for remaining life, coverage for bands).

☐ Detection: **event-level** precision and recall. **No accuracy figure.**

☐ Remaining life: a **window**, never a single date.

☐ If the simple method won or tied, I **recommend it**.

☐ **Deployed** on the training server (dashboard or scheduled report). The README says how to run it.

☐ Repository: code + `requirements.txt` + `README.md`. **No raw NIGCOMSAT data.**

☐ Report: 2–4 pages, the 4 sentences, 3 limitations, the AI-use note.

☐ Demo rehearsed once with a timer: under 5:00.

When things go wrong: error → fix

You see	It means	Fix
`✍️ Fill in the blank first`	A `___` is still there	Replace it, run the cell again
`✍️ Finish the task that makes … first`	You skipped an earlier task	Do that task, then come back
`NameError: name 'hygiene' is not defined`	You skipped the ▶️ toolkit cell	Run it (and STEP 0)
`❌ not a split by time: training reaches …`	The split is shuffled, or the dates are the wrong way round	Train = **before** the cut, test = **from** the cut
`❌ … forecasts but the test has … weeks`	Two methods were scored on different periods	Forecast the same weeks for every method
`ValueError: cannot reindex on an axis with duplicate labels`	`.asfreq()` on data with a repeated timestamp	Drop duplicates first: `df[~df.index.duplicated()]`
`ValueError: Invalid frequency: H`	The old hourly code (pandas 3 removed it)	Use `"h"` (and `"D"`, `"W-MON"`, `"MS"`)
Demo: the dashboard won't load	Server slot, VPN or port	Switch to your screenshot and keep talking; fix it after
Demo: you're at 4:30 and not at the recommendation	Too long on the data or method	Skip to the 4 sentences. Always.
Everything is weird	–	**Runtime → Restart session and run all**

Rules of thumb you'll keep forever

1. **Name the process and its owner.** A tool without an owner is a demo, not a deployment.
2. **Clean, then say what you cleaned.** Every gap filled is an assumption written down.
3. **Split by time, assert it, and score everything on the same period with the same scorer.**
4. **A range, checked.** A band is only as honest as its coverage on weeks it never saw.
5. **If the simple method wins, recommend it.** That's a respected result, not a failure.
6. **Four sentences, one number, one action.** Then stop talking.

🏁 **Finished?** Run `scoreboard()`. Then rehearse your own 5-minute demo with a partner and a timer.